composition. The dependence of the ionization on the atomic composition is a consequence of the different ionization potentials of the different types of atoms of the gas and the different probabilities for electron capture and collision. The ion current is collected on an electrode and measured. This current is a function of the gas pressure and velocity since the higher the pressure, the more ions form, while at higher velocity, the fewer ions are collected as more ions are removed by the gas prior to collection. Such ionization instruments are used in gas chromatographs and other instruments as well as in smoke detection systems (the normal radiation source is ^{241}Am, usually $\leq$40 kBq), where secondary electrons condense on smoke particles, leading to lower mobility for the electrons and a decreased ion current.

Both α- and β-emitters are used in luminescent paint. The fluorescent material is usually ZnS. T and ^{14}C are preferred sources since their β-energies are low, but ^{85}Kr, ^{90}Sr, and ^{147}Pm are also used. The amount or radioactivity varies, depending on the need (watches, aircraft instruments, etc.) but it is usually <400 MBq (<10 mCi), although larger light panels may require >50 GBq (several Curies). For such high activities only T or ^{85}Kr are acceptable because of their relatively low radiotoxicity.

8.13. EXERCISES

8.1. How many ion pairs are produced in 10 m of air of STP by one (a) 5 MeV α-particle, (b) 1 MeV β-particle, and (c) 1 MeV γ-quantum (μ_m(air) ≈ μ_m(water))?

8.2. Estimate the fraction of energy lost through bremsstrahlung for a β-emission of $E_{max} = 2.3$ MeV, when absorbed in aluminum. The effective β-energy must be taken into account.

8.3. A freshly prepared small source of ^{24}Ne had a measured decay rate of 1 GBq 1 s after its preparation. The source is shielded by 10 cm of Pb. ^{24}Ne emits γ-rays; 8% with 0.878 MeV and 100% with 0.472 MeV. Estimate the total integrated dose received at 2 m distance during its life-time after preparation. Neglect build-up factors.

8.4. An acidic aqueous solution is irradiated by α-particles from dissolved ^{239}Pu at a concentration of 0.03 M. The plutonium is originally in its hexavalent state, but is reduced to the tetravalent state by the reaction Pu(VI) + 2H· → Pu(IV) + 2H$^+$. How much of the plutonium can be reduced in one week?

8.5. An acid solution of fresh fission products contains 0.8 g l^{-1} cerium as Ce^{4+}. The γ-flux in the solution corresponds to 520 GBq l^{-1} of an average energy of 0.7 MeV. If half of the γ-flux is absorbed in the solution, what fraction of cerium is reduced to Ce^{3+} in 24 h? Assume the same *G*-value as in Figure 8.7.

8.6. Estimate the average *LET* value in water for β-particles from ^{90}Sr and ^{90}Y, and in Al for T.

8.7. A ^{60}Co irradiation source is calibrated by the Fricke dosimeter for which the *G*-value is assumed to be 1.62 μmol/J. Before the irradiation the optical density *D* of

the solution at 305 nm was 0.049 in a 1 cm cuvette. After exactly 2 h the D had changed to 0.213. Calculate the dose rate when the molar extinction of Fe^{3+} is 217.5 $m^2\ mol^{-1}$.

8.8. A direct reading condenser chamber (pen dosimeter) is charged from a battery pack so that full scale (100) is obtained at 20 V. When completely discharged, the accumulated dose is 5.00 mGy. The gas volume is 4 cm^{-3} air (STP). What is the capacitance of the condenser chamber?

8.14. LITERATURE

S. C. Lind, *Radiation Chemistry of Gases*, Reinhold, 1961.
A. O. Allen, *The Radiation Chemistry of Water and Aqueous Solutions*, D. van Nostrand, 1961.
M. Haissinsky and M. Magat, *Radiolytic Yields*, Pergamon Press, Oxford, 1963.
R. O. Bolt and J. G. Carroll, *Radiation Effects on Organic Materials*, Academic Press, 1963.
B. T. Kelly, *Irradiation Damage to Solids*, Pergamon Press, Oxford, 1966.
J. L. Dye, The solvated electron, *Sci. Am.*, Febr. 1967.
J. R. Cameron, N. Suntharalingham and G. N. Kenney, *Thermoluminescent Dosimetry*, University of Wisconsin Press, 1968.
E. J. Hart (Ed.), *Radiation Chemistry*, Advances in Chemistry Series 81. Am. Chem. Soc., Washington DC 1968.
E. J. Henley and E. R. Johnson, *The Chemistry and Physics of High Energy Reactions*, University Press, Washington DC 1969.
W. Schnabel and J. Wendenburg, *Einführung in die Strahlenchemie*, Verlag-Chemie, 1969.
N. E. Holm and R. J. Berry (Eds.), *Manual on Radiation Dosimetry*, Marcel Dekker, 1970.
J. H. O'Donnell and D. F. Sangster, *Principles of Radiation Chemistry*, Elsevier, 1970.
A. R. Denaro and G. G. Jayson, *Fundamentals of Radiation Chemistry*, Butterworths, 1972.
G. E. Adams, E. M. Fielden, and B. D. Michael (Eds.), *Fast Processes in Radiation Chemistry and Biology*, J. Wiley, 1975.
T. Cairns, *Archeological Dating by Thermoluminescence*, *Anal. Chem.* **48** (1976) 267A.
G. V. Buxton and R. M. Sellers, The radiation chemistry of metal ions in aqueous solutions, *Coord. Chem. Rev.* **22** (1977) 195.
J. Foeldiak (Ed.), *Radiation Chemistry of Hydrocarbons*, Elsevier, 1981.
Farhataziz and M. A. J. Rorgers (Eds.), *Radiation Chemistry, Principles and Applications*, VCH Press Publ. Co, Weinheim 1987.
G. Freeman (Ed.), *Kinetics of Nonhomogeneous Processes*, J. Wiley, 1987.
J. Bednár, *Theoretical Foundations of Radiation Chemistry*, Kluwer Academic Publishers, 1989.
J. R. Greening, *Fundamentals of Radiation Dosimetry*, Adam Hilger, 1989.
V. K. Milinchuk and V.I. Pupikov (Eds.), *Organic Radiation Chemistry Handbook*, Ellis Horwood/J. Wiley, 1989.
J. W. T. Spinks and R. I. Woods, *An Introduction to Radiation Chemistry*, 3rd edn., J. Wiley, 1990.
Y. Tabata (Ed.), *Pulse Radiolysis*, CRC Press Inc., 1991.
D. W. Clegg and A. A. Collyer (Eds.), *Irradiation Effects on Polymers*, Elsevier, 1991.
The Journal of Radiation Physics and Chemistry (Elsevier) regularly contains survey papers of various fields of radiation chemistry.